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B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2024
Sixth Semester

21CSS301T – FULL STACK DEVELOPMENT
(For the candidates admitted from the academic year 2022-2023 onwards)

- Note:**
- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
 - (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours Max. Marks: 75

PART – A (20 × 1 = 20Marks)
Answer ALL Questions

Marks BL CO PO

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|--|------------------|
| 1. Field is a name-value pair in a _____. | 1 2 5 1 |
| (A) Attribute (B) Table | |
| (C) Document (D) Data | |
| | |
| 2. MongoDB stores document in _____. | 1 1 5 1 |
| (A) Store (B) Collections | |
| (C) Indexes (D) Array | |
| | |
| 3. Aggregation operation in MongoDB _____ values from multiple documents. | 1 1 5 1 |
| (A) Dataset (B) Set | |
| (C) Group (D) Record | |
| | |
| 4. Which of the following is the data type in MongoDB? | 1 1 5 1 |
| (A) Time (B) Object | |
| (C) Symbolic (D) Int | |
| | |
| 5. The React.JS written in _____ language. | 1 1 2 1 |
| (A) Python (B) JavaScript | |
| (C) Java (D) Php | |
| | |
| 6. The _____ component must have for every React.JS. | 1 1 2 1 |
| (A) Render component (B) Set initial component | |
| (C) Init render (D) Create Dom | |
| | |
| 7. Choose the method with refers to the parent class in React.JS? | 1 1 2 1 |
| (A) This () (B) Super () | |
| (C) Inherits () (D) Self () | |
| | |
| 8. Identify function which is called to render HTML to a webpage in react. | 1 1 1 1 |
| (A) ReactDOM_Render () (B) Render () | |
| (C) Render_Dom () (D) Dom.HTML () | |

9. A class is a type of function, but instead of using the keyword function to initiate it, which keyword do we use? 1 2 2 1
 (A) Constructor (B) Class
 (C) Object (D) Data object
10. What is a state in react? 1 1 2 1
 (A) A permanent storage (B) Internal storage of the component
 (C) External storage of the component (D) A temporary storage component
11. Which of the following are two ways to handle data in react? 1 1 2 1
 (A) Services, components (B) State, props
 (C) State, services (D) State components
12. What is the full form of REPL? 1 1 3 1
 (A) Read Eval Print Loop (B) Research Eval Program Learn
 (C) Read Earn Point Learn (D) Read Eval Point Loop
13. Default scope in node.JS application is _____. 1 1 3 1
 (A) Global (B) Local
 (C) Local to object (D) Local to class
14. How can we initiate a Node.JS file called hello in Node? 1 1 3 1
 (A) Hello.JS (B) Node Hello.JS
 (C) JavaScript Hello.JS (D) Node hello
15. Which of the following are built-in node modules? 1 1 3 1
 (A) ZLIB (B) HTTPSEC
 (C) Datagram (D) HTTPLIB
16. Which of the following is a valid HTTP method? 1 1 4 1
 (A) Get HTTP (B) Put
 (C) Post HTTP (D) Put HTTP
17. Which of the following node.JS modules is used in file uploads? 1 1 4 1
 (A) Formidable (B) FS
 (C) Express (D) Files
18. Where are the captured values populated regarding the route parameters? 1 1 4 1
 (A) Req.Data (B) App.Locals
 (C) Req.Params (D) App.Global
19. How is it possible to create chainable route handlers for a route path in express.JS? 1 2 4 1
 (A) Using App.Route () (B) Using App.Routes ()
 (C) Using App.Router () (D) Using App.Routing ()
20. Where are the captured values stored in react? 1 1 2 1
 (A) Components state (B) Class
 (C) Function (D) HTML file

PART – B (4 × 10 = 40 Marks)

Answer ANY FOUR Questions

Marks BL CO PO

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| 21. Develop a react form that captures user Email and password. Implement form validation to ensure that the Email follows a valid format and the password meets criteria. Display appropriate error message when validation fails and enable the form submission only then the inputs meets the validation criteria. Provide detailed explanation of your validation approach. | 10 | 3 | 2 | 3 |
| 22. Construct a react application with 3 functional components. Title, description and page. Begin by defining the title component in the title.JS file to render a static title, such as “Welcome to my react app”, subsequently, create the description component in the description.JS file to display a static description “Explore the world of react components”. Finally, compose both the title and description components, implement in the page.JS file. The page component should render a cohesive webpage with a title and description without explicitly passing any data as props between the components. | 10 | 3 | 2 | 3 |
| 23. Create a react functional component named userinfo that accepts two properties username and userrole, from its parent component. Inside the userinfo component, display the provided username and userrole within a paragraph element. This component is designed to be reusable and can dynamically showcase user information based on the data passed to it from the parent component. | 10 | 3 | 2,3 | 3 |
| 24. Explain in simple terms what a “Document” is in MongoDB. How does it differ from a traditional row in a relational database? Briefly describe what a “Collection” is and its purpose in MongoDB. | 10 | 2 | 5 | 3 |
| 25. Develop an Express.JS application that exposes restful end points for a simple resource, e.g “Books”. Create routes to handle get, post, put and delete requests for the “Books” resource. Use an in-memory array as a data store for simplicity, and implement appropriate response for each HTTP method. | 10 | 3 | 3,4 | 3 |
| 26. Explain the concept of the node.JS event loop and how it enables asynchronous operations in node.JS. Discuss the importance of non-blocking I/O operation in the event loop. Provide an example scenario where the event loop enhances the performance of a Node.JS application. | 10 | 2 | 3 | 3 |

PART – C (1 × 15 = 15 Marks)

Answer ANY ONE Question

Marks BL CO PO

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|--|----|---|---|---|
| 27. Build a react application that allows user to explore and search for recipes. The application should include features such as displaying a list of recipes, filtering recipes by category, and viewing detailed information about a selected recipe.
Requirements:
(i) Recipe list components <ul style="list-style-type: none"> • Create a react component (recipe list) to display a list of recipes • Populate the list with mock data, including properties like recipe name, category and a brief description | 15 | 4 | 2 | 3 |
|--|----|---|---|---|

- (ii) Category filtering
 - Implement filtering functionality to allow users to filter recipes based on categories. [Ex. main courses, desserts]
 - Provide UI controls, such as buttons, or drop downs to select recipe categories
 - (iii) Recipe details components
 - Create a separate react component (recipe details) to display detailed information about a selected recipe
 - When a user clicks on a recipe in the list, show the details of the recipe, including ingredients and instructions
 - (iv) Search functionality
 - Implement a search bar that allows user to search for recipes by name.
28. You are asked with building a react application that interacts with a MongoDB database. The application should allow user to retrieve and display data from the database and provide functionality to add new records. Follow the instructions below:
- 15 4 2 3
- (i) Server-side setup
 - Setup a Node.JS server using express to handle backend operations
 - Connect the server to a MongoDB database using a suitable MongoDB Node.JS driver
 - Create a MongoDB schema for the data you will be storing
 - (ii) Read from MongoDB
 - Develop a react component (Data display) that retrieves data from the MongoDB database using an API endpoint on the server
 - Display the retrieved data in a visually appealing manner within the data display component
 - (iii) Write to MongoDB
 - Create a react component (Data entry form) that allows user to input new data
 - Implement functionality within data entry form to send the entered data to the server, which in turn, stores it in the MongoDB database.

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